

Nothing New Under the Sun

Outcome Compression, Realization Compression, and a Surprisingly Long Entropic Detour

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Abstract

This note is a retrospective rather than a priority claim.

We initially approached a family of controlled stochastic processes through several Shannon-information quantities and became interested in the gap between two of them: the minimum entropy of an arbitrary common generator of all policy-indexed transcript laws, and the minimum entropy of one common source that admits a single counterfactually coherent causal realization.

The distinction is mathematically clean. It is also surrounded, on almost every side, by substantial prior work: minimum-entropy coupling, minimum-entropy functional representations, random-map realizations, innovation representations, response-function structural causal models, entropy-constrained hidden-variable models, realization-equivalent mixed strategies, positive realizations, and sequential channel synthesis.

Our initial literature search was not broad enough. We apologize for the earlier overstatement of novelty. After repeatedly finding closer predecessors under different terminology, we no longer claim that any individual ingredient, formulation, or theorem in this note is new. Some specific combinations may or may not have appeared verbatim; we have not established priority, and we deliberately make no such claim.

What remains is a useful calculation, a sharp conceptual distinction, and a record of an unexpectedly instructive route through several neighboring theories. We also report one failed strengthening: a finite-horizon linear entropy gap does not naively upgrade to a stationary ergodic rate theorem under full-history response-tree semantics. The failure exposes an important structural fact: after two deterministic policy histories differ in action, they never re-merge as the same history–action node.

1 A correction of scope

The investigation gradually converged on two quantities associated with the same family of policy-indexed transcript laws

$$\{\mu_n^\pi : \pi \in G\}.$$

The first is

$$\kappa_n^G := \min_{P \in \mathcal{C}(\mu_n^\pi : \pi \in G)} H_P((T_n^\pi)_{\pi \in G}),$$

the ordinary multi-marginal minimum-entropy coupling value.

The second restricts the same coupling problem to tuples that can arise from one deterministic counterfactual response object:

$$\chi_n^G := \min_{\substack{P \in \mathcal{C}(\mu_n^\pi : \pi \in G) \\ \text{supp } P \subseteq \text{TC}_n(G)}} H_P((T_n^\pi)_{\pi \in G}).$$

Their difference is

$$\Gamma_n^G := \chi_n^G - \kappa_n^G \geq 0.$$

At first sight this appeared to isolate a new information-theoretic resource. A broader literature search made that interpretation increasingly difficult to defend.

Minimum-entropy coupling is established [1, 2]. Minimum-entropy exogenous and functional representations are closely tied to the same coupling problem [3, 4]. Low-entropy innovation representations of stochastic processes are established [5]. Random-map realizations of Markov chains and their entropy redundancy were studied earlier [6]. Response-function structural causal models and counterfactually consistent joint SCMs are established [7]. Extensive-form game theory has long studied realization-equivalent mixtures and compression of their support [8]. Device-independent work has directly bounded or optimized the Shannon entropy of messages and shared hidden variables needed to reproduce observed behavior [9, 10]. Positive-realization theory asks for compact stochastic mechanisms producing a prescribed process [11]. Sequential channel synthesis studies randomness and communication resources for causal generation of stochastic channels [12].

Accordingly, we withdraw every broad novelty claim of the form

“a coherent realization can require more informational resources.”

That theme has many predecessors.

We retain the notation $\kappa_n, \chi_n, \Gamma_n$ only because it makes one particular comparison transparent.

2 The calculation that motivated the detour

Consider the two-stage crossed-policy construction.

At the first stage,

$$Y \sim \text{Bern}(1/2).$$

At the second stage the controller chooses $A \in \{0, 1\}$. Let α and β be arbitrary finite output laws, independent of Y , with

$$\mathcal{L}(Z \mid A = 0) = \alpha, \quad \mathcal{L}(Z \mid A = 1) = \beta.$$

Take the two policies

$$\pi(Y) = Y, \quad \sigma(Y) = 1 - Y.$$

The two complete transcript laws have the same multiset of atom probabilities. Therefore

$$\kappa_2 = 1 + \frac{H(\alpha) + H(\beta)}{2}.$$

A single response realization must preserve the same value of the shared first observation Y under both policy coordinates. Conditional on either value of Y , it must jointly realize both second-stage action laws. Hence

$$\chi_2 = 1 + \text{MEC}(\alpha, \beta),$$

and therefore

$$\boxed{\Gamma_2 = \text{MEC}(\alpha, \beta) - \frac{H(\alpha) + H(\beta)}{2}.} \tag{1}$$

For example, with

$$\alpha = \delta_0, \quad \beta = \text{Unif}\{1, \dots, m\},$$

one obtains

$$\kappa_2 = 1 + \frac{1}{2} \log_2 m,$$

$$\chi_2 = 1 + \log_2 m,$$

and

$$\Gamma_2 = \frac{1}{2} \log_2 m. \quad (2)$$

Equation (1) is useful, but we make no claim that the information-theoretic object on its right-hand side is new. It is built entirely from established minimum-entropy coupling machinery. Its value for us was interpretive: it made precise the difference between allowing policy-dependent relabelling of a source and requiring the source to represent one common causal object.

3 A finite-horizon linear construction

A more striking finite-horizon calculation is possible with fixed binary alphabets.

Let

$$\mathcal{A} = \mathcal{O} = \{0, 1\}.$$

At the first stage the world emits

$$Y \sim \text{Bern}(1/2).$$

The two policies then choose crossed actions

$$A_2^\pi = Y, \quad A_2^\sigma = 1 - Y.$$

If $A_2 = 0$, the remaining $n - 1$ outputs are independent fair bits. If $A_2 = 1$, the remaining outputs are deterministic zeros.

The two policy transcript laws differ only by exchanging the labels of the two values of Y . Consequently

$$\kappa_n = 1 + \frac{n-1}{2} = \frac{n+1}{2}.$$

A common realization must keep the first-stage value Y fixed across the two policy coordinates. Conditional on Y , exactly one policy requires the full fair-bit tail, so

$$\chi_n = 1 + (n-1) = n.$$

Thus

$$\boxed{\Gamma_n = \frac{n-1}{2}}. \quad (3)$$

This is an elementary linear finite-horizon separation. We no longer attach a novelty claim to it. The more interesting question for the present retrospective was whether the same mechanism could be turned into a stationary ergodic entropy-rate separation.

4 The stationary-ergodic attack

The obvious idea is to repeat the two-stage gadget forever. That idea fails for a structural reason.

Lemma 1 (No re-merging of full histories). *Let two deterministic policy coordinates follow complete histories*

$$h_t^\pi \quad \text{and} \quad h_t^\sigma.$$

If at some time τ their actions differ,

$$A_\tau^\pi \neq A_\tau^\sigma,$$

then

$$h_t^\pi \neq h_t^\sigma \quad \text{for every } t \geq \tau.$$

Consequently, after the first action divergence, the two coordinates never again visit the same full history-action node.

Proof. The action A_τ is part of every later full history. Since the two values differ, the later history strings differ permanently. $\square$

This elementary observation matters because response-tree consistency is defined on *full history-action nodes*. A physical Markov state may later reset to the same value under two policies, but their response-tree histories have not reset.

Therefore a regenerative construction does not concatenate copies of the finite-horizon premium in the naive way. Once the policies have diverged in action, later physical resets do not create new shared response-tree nodes.

4.1 Why the linear finite-horizon example works

The linear gap in (3) does not arise from repeatedly creating new shared nodes. It uses only one shared prefix.

The common bit Y decides which policy receives a high-entropy tail. An unconstrained coupling is free to pair

$$(\pi, Y = 0)$$

with

$$(\sigma, Y = 1),$$

thereby aligning the two high-entropy tails on the same source atoms. A response-tree coupling may not perform that cross-pairing because the first observation must agree samplewise:

$$Y^\pi = Y^\sigma.$$

The single one-bit prefix therefore locks the coupling of an arbitrarily long tail.

The infinite extension of this construction has an obvious defect. The initial selector chooses a persistent “random-tail” versus “deterministic-tail” regime. The resulting natural stationary mixture contains a permanent latent component and is not ergodic.

4.2 Why the obvious ergodic repair fails

One can try to repair the construction by allowing the regime to reset ergodically. But after the first policy action divergence, each reset occurs at two different full histories. By the no-re-merging lemma, response-tree consistency does not require those later reset variables to be represented by the same node or by the same counterfactual response.

Thus the reset destroys the permanent nonergodic regime, but simultaneously destroys the mechanism that made the long tail expensive.

We therefore did *not* obtain a stationary ergodic theorem of the form

$$\liminf_{n \rightarrow \infty} \frac{\chi_n - \kappa_n}{n} > 0$$

for a fixed finite-state controlled process and fixed finite policy family under the full-history response-tree definition.

This should not be read as an impossibility theorem. We have not proved one. It is simply where the direct construction breaks.

Remark 1. *A different realization semantics could change the conclusion. If consistency were imposed on a Markov state rather than on the complete history, two policy trajectories could genuinely re-merge after a state reset. That would be a different model, closer to finite-state realization theory and computational mechanics, and we do not pursue it here.*

5 What survived

After the literature search and the failed stationary-ergodic upgrade, the most defensible statement is not a claim of discovery.

It is the following distinction:

outcome compression:	$\min H(\text{one source reproducing each policy law}),$
realization compression:	$\min H(\text{one source representing one coherent causal object}).$

These two optimization contracts need not coincide in the examples above. That observation remains useful as bookkeeping even if every ingredient, and perhaps every consequence, already exists somewhere in the literature under another name.

The important correction is that we no longer equate “we did not initially recognize the antecedent” with “there is no antecedent.”

6 Epilogue

This project began with a broad intuition about the informational price of a single counterfactual world. It then passed through minimum-entropy couplings, functional representations, response functions, causal marginal problems, contextual gluing, random mappings, extensive-form games, hidden variables, positive realizations, computational mechanics, and sequential channel synthesis.

At several points an apparently new object turned out to be an old object in unfamiliar notation. At several others, a promising generalization collapsed when the operational contract was written carefully enough.

That is not a failure of the exercise.

The theorem survived longer than the novelty claim. The novelty claim, in turn, survived only until the literature search became sufficiently broad. What remained was a much cleaner understanding of the distinction being measured, why the finite-horizon examples work, and why the most obvious stationary extension does not.

We therefore end with no claim of priority.

No new continent was discovered.

The map became considerably better.

And, for what it is worth, it was a fascinating ride.

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